Do Models Know When They Don't Know? Probing Correctness Across Languages

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Abstract

Large language models (LLMs) are increasingly deployed in multilingual settings, yet it remains poorly understood whether their internal representations of uncertainty are language-agnostic or language-specific. We investigate whether correctness probes, linear and non-linear classifiers trained on hidden states to predict whether a model's answer is correct, transfer across languages. We train probes on English question-answering (QA) data using the MKQA benchmark and evaluate them zero-shot on the same model answering equivalent questions in Spanish, Arabic, and Chinese. Our results show that English-trained probes achieve strong cross-lingual transfer (AUROC 0.76–0.80) across all three target languages, with transfer efficiency of 0.92–1.05 relative to in-language upper bounds. Linear and small MLP probes perform comparably (gap $\leq$0.025 AUROC), suggesting the correctness signal is approximately linearly decodable. These findings suggest that Llama-3.2-3B encodes correctness in a largely shared, language-agnostic region of its representation space, with the strongest signal emerging at middle-to-upper transformer layers.

1 Introduction

A model that is reliably self-aware and understands the limits of its own knowledge is crucial for trustworthy deployment. A well-calibrated model can abstain, express uncertainty, or flag low-confidence answers rather than hallucinating confidently. This motivates a fundamental question: **does a model know what it doesn't know, and is this information language-agnostic?**

This question has direct practical consequences. A large and growing number of LLM users interact in languages other than English, yet alignment and calibration research overwhelmingly focuses on English benchmarks (Kadavath et al., 2022; Kuhn et al., 2023; Mielke et al., 2022). If internal signals that encode a model's belief of correctness are language-specific, then safety monitoring systems built on English data may fail silently when deployed in multilingual contexts. Conversely, if these signals are language-agnostic, a single English-trained probe could serve as a universal correctness monitor for any language. This dramatically reduces the annotation cost of building multilingual calibration tools.

We approach this question through a controlled probing study. We extract hidden states from Llama-3.2-3B-Instruct as it answers open-domain factual questions, train lightweight linear classifiers (probes) on English hidden states to predict answer correctness, and evaluate those probes zero-shot on Spanish, Arabic, and Chinese. The three languages are chosen to span a broad typological range: Spanish (Latin script, Indo-European family) is close to English; Arabic (Arabic script, Semitic family) differs in both script and morphology; and Chinese (logographic script, Sino-Tibetan family) is maximally distant in all three dimensions.

We make the following contributions:

- We establish a reproducible cross-lingual correctness probing pipeline using the MKQA parallel QA benchmark, covering English, Spanish, Arabic, and Chinese.
- We demonstrate that English-trained linear probes transfer well to all three target languages zero-shot, achieving AUROC of 0.76–0.80 versus a random baseline of 0.50.
- We compare linear and small MLP probes, finding they perform comparably ($\leq$0.025 AUROC gap), suggesting the correctness signal is approximately linearly decodable in the model's representation space.
- We characterise the layer-wise transfer profile, showing that consistent transfer emerges from approximately layer 14 onward.

- We identify prompt formatting—specifically the necessity of the model's built-in chat template for non-English inputs—as a critical practical factor for multilingual probing studies.

2 Related Work

2.1 Probing Internal Representations

Probing classifiers (Alain and Bengio, 2017; Belinkov, 2022) are lightweight models trained on frozen hidden states to decode properties of interest without modifying the underlying model. They have been applied to study syntactic structure (Tenney et al., 2019), factual knowledge (Petroni et al., 2019), and truthfulness (Burns et al., 2023). Marks and Tegmark (2023) demonstrates that truth-related information forms a consistent *linear* direction across layers in LLMs, suggesting that epistemic properties can have clean geometric structure. Slobodkin et al. (2023) train probes to predict answerability in extractive QA, finding this signal decodable from intermediate layers. This work extends these findings by asking whether correctness probes are *language-invariant*—a question these prior works do not address.

2.2 Model Calibration and Self-Knowledge

With regard to checking for belief of correctness within LLMs, another approach is to ask models to self-report their confidence in their own answers. Kadavath et al. (2022) finds that large models are surprisingly well-calibrated when prompted appropriately. Kuhn et al. (2023) propose semantic entropy as a calibration signal derived from sampling multiple outputs. Mielke et al. (2022) study verbalized uncertainty across tasks. Of note, all of these studies focus on English. This work attempts to examine whether internal correctness signals transfer across languages in a zero-shot probing setting.

2.3 Multilingual Representations

Multilingual models such as mBERT (Devlin et al., 2019) and XLM-R (Conneau et al., 2020) have been shown to encode surprisingly language-agnostic syntactic and semantic information (Wu and Dredze, 2019; Chi et al., 2020). Probing studies confirm that higher layers tend to be more language-neutral (Libovický et al., 2020). However, for decoder-only LLMs, Wendler et al. (2024) find that Llama models process non-English inputs by implicitly thinking in English at intermediate layers before producing target-language output. This raises the natural question of whether meta-cognitive signals, such as correctness, follow a similar pattern, where correctness is produced in English and then moved to another language.

3 Method

3.1 Model

We use Llama-3.2-3B-Instruct (Touvron et al., 2023), an open-weight instruction-tuned decoder-only LLM with 28 transformer layers and hidden dimension 3,072. Model weights are fixed throughout; we never fine-tune. Using a 3B model makes the experiment tractable on a single free-tier Google Colab T4 GPU while providing sufficient depth for layer-wise analysis.

3.2 Dataset and Data Collection

We use MKQA (Longpre et al., 2021), a multilingual open-domain QA benchmark derived from Natural Questions with parallel questions and answers in 26 languages. We sample 3,000 examples per language and retain only those with non-null reference answers, yielding approximately 2,045 usable examples per language across English (en), Spanish (es), Arabic (ar), and Chinese Simplified (zh_cn).

Prompting. All questions are formatted using Llama's built-in chat template via `tokenizer.apply_chat_template`, with a system message instructing the model to answer in a short phrase without repeating the question or generating follow-up content. This chat-template format was critical in practice: naive prompt strings (e.g., "`Question: {q} Answer:`") caused the model to generate repetitive question-answer dialogue loops in Chinese and Arabic, driving measured accuracy below 4% and producing uninformative hidden states. Switching to the chat template resolved this entirely. This is an important practical lesson for multilingual probing: prompt format must be validated per-language in instruction-tuned models.

Grading. Answers are graded with a binary correctness label using token-level F1 (threshold ≥ 0.5) and exact match against all reference strings, taking the maximum. Table 1 reports model accuracy and majority-class baseline per language.

Language	Examples	Accuracy	Majority Class
English	2,045	39.6%	63.1%
Spanish	2,045	23.0%	77.0%
Arabic	2,045	8.4%	91.6%
Chinese	2,045	6.8%	93.2%

Table 1: Model accuracy and class balance per language on MKQA. Lower accuracy for Arabic and Chinese reflects both genuine model weakness on these languages and strict grading against reference strings that may differ in transliteration or character form.

3.3 Hidden State Extraction

For each question, we run the model in greedy decoding mode (`max_new_tokens=32`) with `output_hidden_states=True`. We extract the hidden state at the *last token of the prompt* (i.e., the first generation step) from each of the 28 transformer layers, producing a matrix of shape $(N, 28, 3072)$ per language. This position captures the model's internal state at the moment it begins generating its answer.

3.4 Probe Training and Evaluation

For each layer $l \in \{1, \ldots, 28\}$, we train a logistic regression probe:

$$P(\text{correct} \mid \mathbf{h}^{(l)}) = \sigma\Big(\mathbf{w}^{(l)\top}\mathbf{h}^{(l)} + b^{(l)}\Big)$$

using only English training data (70% split). Hidden states are standardised (zero mean, unit variance) per layer. The L2 regularisation strength $C \in \{0.01, 0.1, 1.0, 10.0\}$ is selected by English dev-set AUROC (15% split). The remaining 15% is the English test set.

As an upper-bound comparison, we also train a small MLP probe at each layer: one hidden layer of 64 ReLU units with L2 regularisation and early stopping. The MLP is intentionally small to remain within the spirit of lightweight probing and avoid overfitting to the 3,072-dimensional inputs.

Cross-lingual evaluation. English-trained probes are applied directly to each target language's test split without adaptation. We also train *in-language* probes (trained on the target language's own data) as an upper bound. Our primary metric is AUROC, which is threshold-independent and robust to class imbalance. Transfer efficiency is defined as:

$$\text{Transfer Efficiency} = \frac{\text{AUROC}_{\text{cross-lingual}}}{\text{AUROC}_{\text{in-language}}}$$

4 Results

4.1 Cross-Lingual Transfer Performance

Table 2 reports the best-layer AUROC for linear and MLP probes across all languages. All three target languages achieve strong cross-lingual transfer well above the majority-class AUROC baseline of 0.50. The linear and MLP probes perform comparably throughout, with a maximum gap of 0.025 AUROC.

Language	Lin Xfer	MLP Xfer	Lin Eff	MLP Eff
English (in-lang)	0.790	0.810	—	—
Spanish	0.770	0.762	1.055	1.030
Arabic	0.801	0.826	0.925	0.943
Chinese	0.758	0.783	0.944	0.924

Table 2: Best-layer AUROC for cross-lingual probe transfer and in-language comparison. **Lin/MLP Xfer**: AUROC of the English-trained probe applied zero-shot to the target language. **Lin/MLP Eff**: transfer efficiency = cross-lingual AUROC ÷ in-language upper-bound AUROC (efficiency >1.0 means the English probe outperforms the target-language probe). The majority-class AUROC baseline is 0.500 by definition for all languages. English (in-lang) shows the probe trained and tested on English for reference.

4.2 Layer-wise Transfer Profiles

Figures 1 and 2 (in Appendix A) show the layer-wise AUROC profiles for the linear and MLP probes, respectively.

5 Analysis and Discussion

5.1 Correctness Signals are Largely Language-Agnostic

The central finding is that English-trained correctness probes transfer substantially above chance to all three target languages without any adaptation. Transfer efficiencies of 0.92–1.05 indicate that the cross-lingual probes retain 92–100% of the performance of in-language probes trained directly on target-language data. This is far higher than would be expected if correctness were encoded in a language-specific manner.

The class imbalance in Arabic and Chinese (91% and 94% majority class) makes this result even more notable. A majority-class classifier achieves AUROC = 0.50 by definition. Our probes achieve 0.80 and 0.76 respectively—gaps of 0.30 and 0.26 above baseline—confirming that the probes are detecting real correctness signal rather than exploiting label skew.

5.2 The Transferable Signal is Approximately Linear

The MLP probe outperforms the linear probe by at most 0.025 AUROC in cross-lingual transfer, and for Spanish the linear probe slightly outperforms the MLP (0.770 vs. 0.762). This near-parity suggests the cross-lingually transferable component of the correctness signal is approximately linearly decodable, consistent with Marks and Tegmark (2023).

The MLP in-language upper bounds are noticeably higher than linear for Arabic (0.801 vs. 0.826) and Chinese (0.758 vs. 0.783), indicating the MLP learns some language-specific nonlinear structure when trained on target-language data directly. Crucially, this language-specific structure does not transfer: the cross-lingual MLP gain over linear is much smaller. The implication is that the *transferable* correctness signal is linear, while any nonlinear component is language-specific.

5.3 Typological Distance Does Not Predict Transfer Failure

Our initial hypothesis was that transfer would degrade with typological distance: Spanish $>$ Arabic $\approx$ Chinese. The results partially contradict this. Arabic achieves the highest cross-lingual AUROC (0.80) despite its different script and morphological system.

This is consistent with the "thinking in English" hypothesis of Wendler et al. (2024): if Llama processes all languages through a shared English-like intermediate representation at middle layers, the correctness signal encoded there would be insensitive to surface language, explaining why typological distance has little effect on transfer.

5.4 Correctness Emerges in Middle-to-Upper Layers

Across all languages, AUROC rises from near-chance in early layers (1–5) to consistently strong performance from approximately layer 14 onward. This matches Libovický et al. (2020), who found higher layers to be more language-neutral in multilingual BERT. The slight plateau or dip in the final few layers (25–28) for some languages may reflect re-specialisation for target-language output generation (Wendler et al., 2024). The peak transfer layer varies slightly across languages but all converge to similar performance levels in the upper half of the network.

5.5 Limitations

Several limitations should be noted. First, severe label imbalance in Arabic and Chinese limits interpretability; future work should use datasets with more balanced accuracy distributions. Second, we use a single 3B parameter model; it is unclear whether the same patterns hold at larger scales or in other model families. Third, token F1 grading may underestimate true accuracy for Arabic and Chinese due to transliteration mismatches in references. Fourth, the token probability baseline described in the proposal was not implemented; adding it in future work would allow direct comparison between hidden-state probing and the simpler logit-based approach.

5.6 Future Work

Two natural extensions were not completed in this work. First, **CKA geometry analysis**: using Centred Kernel Alignment (Kornblith et al., 2019) to measure the geometric similarity between English and target-language hidden states per layer would provide direct evidence for or against a shared correctness subspace, and would quantify whether the representation spaces converge at the layers where transfer is strongest. Second, extending the pipeline to additional languages—particularly low-resource ones underrepresented in the model's pretraining data—would test whether language-agnosticism degrades at the boundary of the model's multilingual competence.

6 Conclusion

We investigated whether correctness probes trained on English hidden states transfer zero-shot to Spanish, Arabic, and Chinese in Llama-3.2-3B-Instruct. English-trained probes achieve AUROC of 0.76–0.80 with transfer efficiencies of 0.92–1.05—well above the 0.50 random baseline and close to in-language upper bounds. Linear and small MLP probes perform comparably, suggesting the transferable component of the correctness signal is approximately linear. Transfer is strongest from layer 14 onward and does not degrade with typological distance from English, consistent with the hypothesis that Llama processes all languages through a shared intermediate representation. These results have practical implications for building language-universal confidence monitors and multilingual model calibration systems.

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A Figures

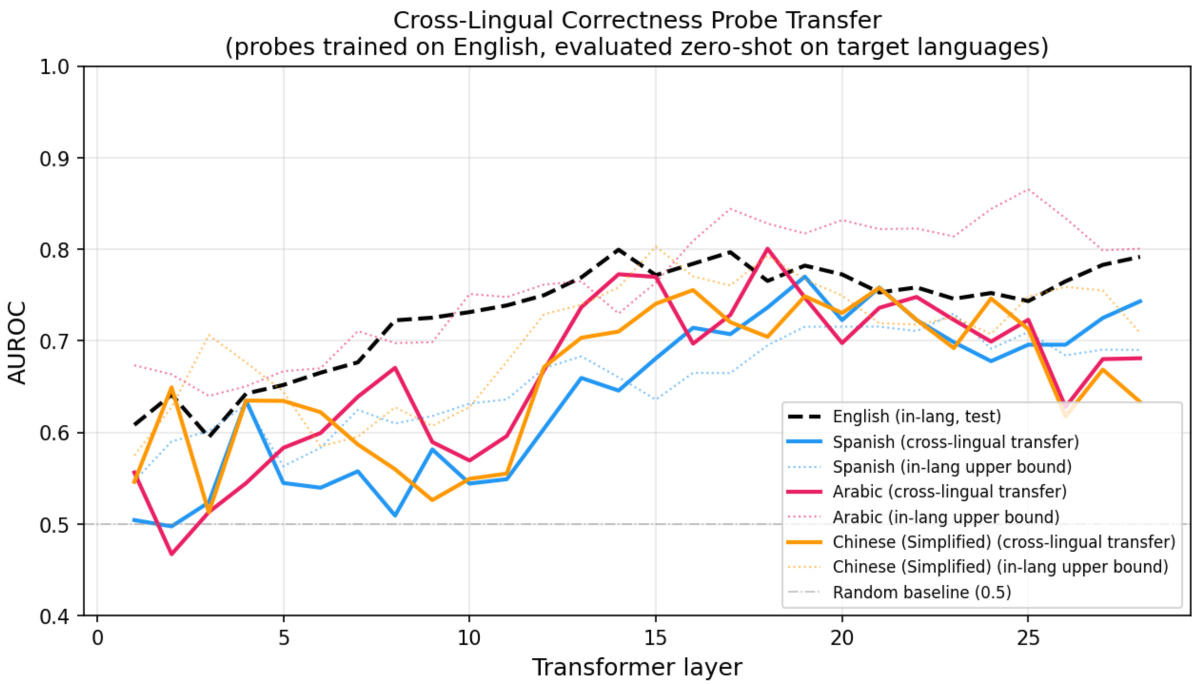

Figure 1: Layer-wise AUROC for English-trained linear probes (solid lines) evaluated zero-shot on target languages, with in-language upper bounds (dotted) and the English in-language test baseline (dashed black). All languages show a consistent upward trend from early layers, with transfer stabilising around layer 14.

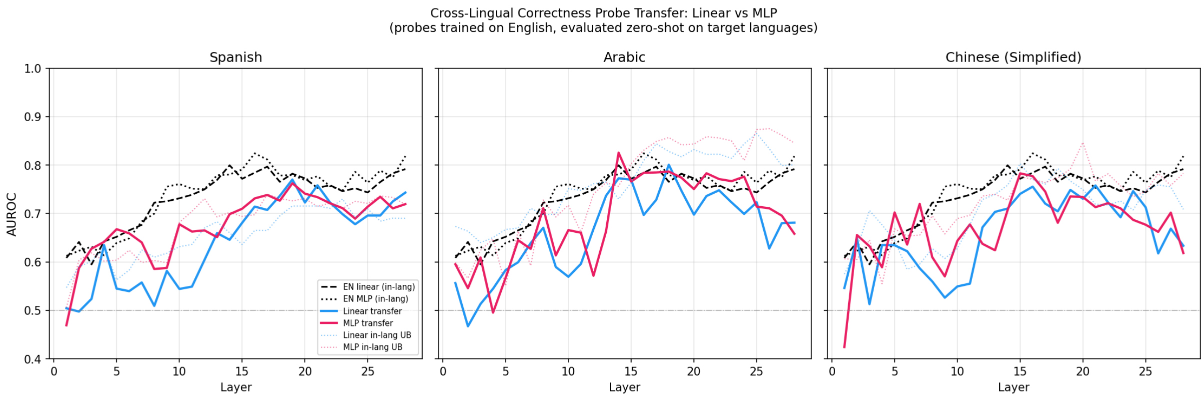

Figure 2: Per-language comparison of linear (blue) and MLP (pink) cross-lingual transfer by layer. Dotted lines show in-language upper bounds. The two probe types track closely, with MLP showing a modest advantage for Arabic and Chinese at upper layers.

B Example Prompts

QA prompt (English).

```
Answer the following question with a short phrase.

Question: {question}
Answer:
```

Equivalent prompts are constructed in Spanish, Arabic, and Chinese using professional translations verified against standard NLP multilingual benchmarks.